Experiment 8: Depth First Search (DFS)

Aim

To write a Python program to implement the Depth First Search (DFS) algorithm.

Procedure

1. Create a graph using an adjacency list.
2. Select the starting node.
3. Mark the node as visited.
4. Display the node.
5. Recursively visit each unvisited neighboring node.
6. Continue until all reachable nodes are visited.

Source Code

main.py

Visualize

Run ▶

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            dfs(neighbour)

start = input("Enter starting node: ")

print("DFS Traversal:")
dfs(start)
```

Output

```
Enter starting node: A
DFS Traversal:
A B D E F C
```

Result

Thus, the Depth First Search (DFS) algorithm was implemented successfully.